

Academic Programs

B.Tech (Mechanical Engineering)

Programme Educational Objectives (PEOs)

PEO 1: To prepare students for successful careers as mechanical

engineers in

organizations

that meet the needs of Indian and global/multinational industrial/research

establishments.

PEO 2: To provide a strong foundation in mathematical,

scientific and engineering

fundamentals in both domain and cross domain spheres, that enables students to

visualize,

analyze and solve mechanical engineering problems and be innovative and research

oriented.

PEO 3: To train students with a wide spectrum of scientific and

Mechanical

engineering

courses so that students could comprehend, analyze, design and create products

and services,

which promotes entrepreneurship culture in the campus that address real life

problems,

which are efficient and cost effective.

PEO 4: To inculcate professional and ethical attitude in

students, impart effective

communication skills and ability to work in teams with multidisciplinary

approach, be part of

and interact with professional bodies so as to resolve engineering issues of

social relevance.

PEO 5: To provide students with an academic environment that

fosters excellence,

leadership,

yearning to pursue higher studies and passion for lifelong learning so as to

have a successful

professional career.

Program Specific Outcomes (PSOs):

PSO 1: Apply expertise in design, manufacturing, materials and

energy systems to

develop

innovative and sustainable solutions that drive industrial growth, ensure

safety, promote

societal well-being, and support green practices.

PSO 2: Create innovative research, develop sustainable

products, formulate

Mechanical Engineering

managerial

and entrepreneurial strategies to implement emerging technologies for addressing industrial, societal, and global challenges.

Program Outcomes (POs):

PO1: Engineering Knowledge: Apply knowledge of mathematics,

natural science,

computing,

engineering fundamentals and an engineering specialization as specified in WK1 to WK4

respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research

literature and analyze

complex

engineering problems reaching substantiated conclusions with consideration for sustainable

development. (WK1 to WK4)

PO3: Design/Development of Solutions:

Design creative solutions for complex engineering

problems and design/develop systems/components/processes to meet identified needs with

consideration for the public health and safety, whole-life cost, net zero carbon, culture, society

and

environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems:

Conduct investigations of complex

engineering problems using research-based knowledge including design of experiments, modelling,

analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply

appropriate techniques,

resources and

modern engineering & IT tools, including prediction and modelling recognizing their limitations

to

solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal

and environmental

aspects

while solving complex engineering problems for its impact on sustainability with reference to

economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics:

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Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function

effectively as an individual,
and as a
member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively

within the engineering
community
and society at large, such as being able to comprehend and write effective reports and design
documentation, make effective presentations considering cultural, language, and learning
differences.

PO10: Project Management and Finance: Apply knowledge and

understanding of
engineering
management principles and economic decision-making and apply these to one's own
work, as a
member and leader in a team, and to manage projects and in multidisciplinary
environments.

PO11: Life-Long Learning: Recognize the need for, and have

the preparation and
ability for
i) independent and life-long learning ii) adaptability to new and emerging
technologies and
iii)
critical thinking in the broadest context of technological change. (WK8)

M.Tech. (Advanced Manufacturing Systems)

PEO 1: To pursue professional career in manufacturing domain

meeting the needs of
Indian and
global/multinational organizations.

PEO 2: To demonstrate strong foundation in the manufacturing

domain enabling them
to visualize,
analyze and solve practical problems with due consideration for economical,
safety and environmental
concerns

PEO 3: To undertake research and engage in continuous

upgradation of skills in the
manufacturing
domain

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PEO 4: To exhibit leadership qualities with demonstrable
attributes in life-long
learning to contribute
to societal needs

Programme Outcomes (POs)

PO1: An ability to independently carry out research
/investigation and development
work to solve
practical problems

PO2: An ability to write and present a substantial technical
report/document

PO3: An ability to demonstrate a degree of mastery over the
manufacturing systems
and processes.

PO4: Ability to design and develop manufacturing systems with
due consideration for
economic,
safety, and sustainability concerns

PO5: Ability to assimilate advancements, apply engineering
judgment, and use modern
tools to solve
engineering problems responsibly, considering their global impact and ethical
implications.

M.Tech. (CAD/CAM)

PEO-1: To empower the students by providing necessary knowledge
base, critical
thinking and
problem-solving capabilities in the field of Computer Aided Design &
Computer Aided
Manufacturing
(CAD/CAM) and allied fields so that they can excel in their profession, in
industry, higher studies
and
Research & Development.

PEO-2: To develop core competencies in the field of CAD/CAM, so
as to conduct
experiments, comprehend, analyze, design and use appropriate techniques and
tools to provide optimal
solutions for the industry related problems.

PEO-3: To inculcate the responsibility to the society at large
by sensitizing
regulatory and Intellectual Property related issues along with communication
skills and to promote
entrepreneurship with sufficient knowledge of project/ finance management

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techniques for ensuring
their career success.

PEO-4: To motivate the students not only to be excellent in
academics, professional
ethics, team work, leadership skills but also to be lifelong learners in
upcoming technologies for
successful professional career.

PO1: An ability to independently carry out research
/investigation and development
work to solve practical problems.

PO2: An ability to write and present a substantial technical
report/document.

PO3: An ability to demonstrate a degree of mastery over the
CAD/CAM technologies.

PO4: An ability to develop skills in research, experimentation,
and analysis,
enabling contributions to CAD/CAM advancements individually or in teams.

PO5: An ability to imbibe professional ethics and intellectual
integrity in the
consideration of impact of research outcomes for sustainable development of the
society.